

Retail Business Performance & Profitability Analysis

Abstract

This project analyzes the Superstore retail dataset using MySQL and Power BI. The objective was to evaluate sales performance, profitability, customer behavior, category performance, regional trends, and the impact of discounts on profits.

Tools Used

- MySQL Workbench (Data Import & SQL Analysis)
- SQL (Aggregation, Grouping, Sorting, Business Queries)
- Power BI (Dashboard Development & Visualization)
- CSV Dataset (Superstore Retail Data)

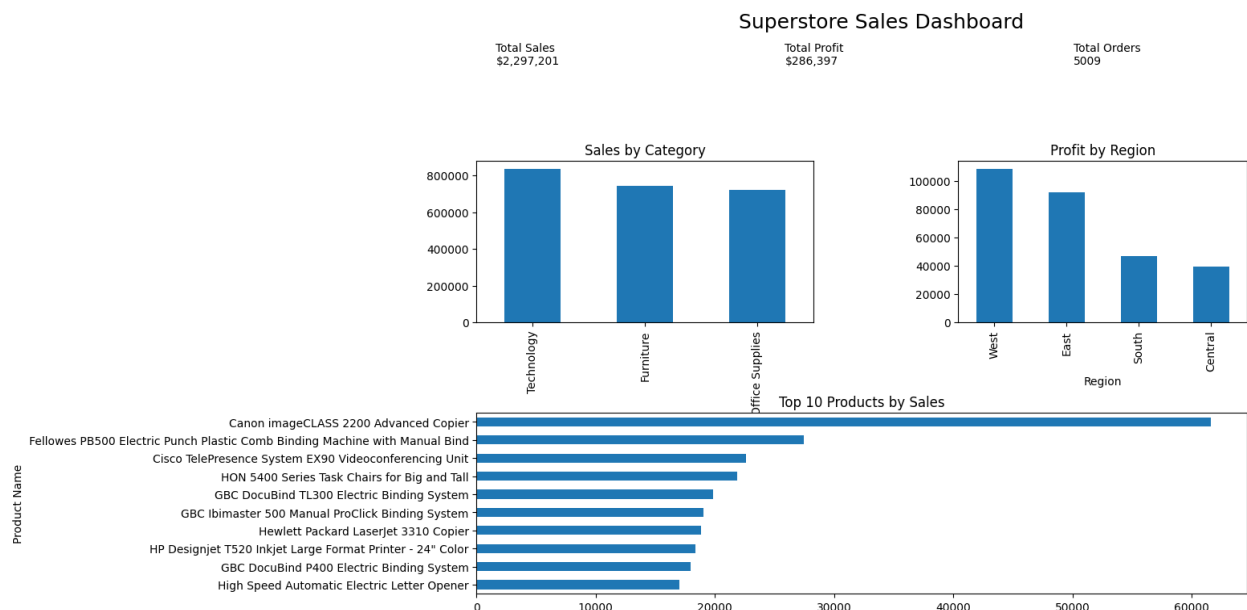
Project Workflow

1. Imported Superstore CSV dataset into MySQL.
2. Performed data validation and record count checks.
3. Executed SQL queries for sales, profit, customer, category and state analysis.
4. Identified top-performing and low-performing regions.
5. Analyzed discount impact on profitability.
6. Built an interactive Power BI dashboard using SQL findings.

SQL Analysis Performed

- Total Sales Analysis
- Total Profit Analysis
- Total Orders Analysis
- State-wise Sales Analysis
- State-wise Profit Analysis
- Loss-making States Identification
- Category-wise Sales & Profit Analysis
- Customer-wise Sales & Profit Analysis
- Discount vs Profit Analysis

Dashboard Overview



Key Findings

- Technology generated the highest sales among categories.
- Western region produced the highest profit.
- A small number of products contributed significantly to revenue.
- Discounts negatively impacted profitability at higher levels.
- Customer and regional performance varied considerably, highlighting opportunities for targeted business strategies.

Conclusion

The project successfully transformed raw retail transaction data into actionable business insights. Using SQL, critical performance metrics were calculated and analyzed. Power BI was then used to visualize these findings through an interactive dashboard. The analysis revealed strong category performance, profitable regions, and the influence of discounts on profit margins. The dashboard can support management decisions related to sales strategy, customer targeting, and profitability improvement.